

MATH Unit 2 Previous Year Questions

Mathematics (MATH) · Unit 2 · Differential Calculus · 2025

Questions collected from previous year university examinations (2021–2025).

Part A — Short answer (2 marks)

1. Explain Mean Value Theorems with a neat diagram and a suitable example. [2025]
2. Discuss the role of Taylor and Maclaurin Series in Differential Calculus. [2023]
3. Write short notes on Partial Derivatives. [2025]
4. What is meant by Jacobians? Discuss its advantages and limitations. [2022]

Part B — Long answer (10 marks)

1. Write short notes on Maxima and Minima. [2025]
2. What is meant by Lagrange Multipliers? Discuss its advantages and limitations. [2025]
3. Discuss the role of Mean Value Theorems in Differential Calculus. [2024]
4. Define Taylor and Maclaurin Series. Derive or justify the key properties involved. [2025]
5. What is meant by Partial Derivatives? Discuss its advantages and limitations. [2022]
6. Discuss the role of Jacobians in Differential Calculus. [2024]
7. Write short notes on Maxima and Minima. [2021]
8. Discuss the role of Lagrange Multipliers in Differential Calculus. [2024]
9. Write short notes on Mean Value Theorems. [2022]
10. Explain Taylor and Maclaurin Series with a neat diagram and a suitable example. [2021]